

2.1.1. Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

Code:

```
list = []
```

```
while True:
```

```
    print("1. Add")
```

```
    print("2. Remove")
```

```
    print("3. Display")
```

```
    print("4. Quit")
```

```
    ch = int(input("Enter choice: "))
```

```
    if ch == 1:
```

```
        num=int(input("Integer: "))
```

```
        list.append(num)
```

```
        print("List after adding:",list)
```

```
    elif ch == 2:
```

```
        if not list:
```

```
            print("List is empty")
```

```
        else:
```

```
            num1=int(input("Integer: "))
```

```
            if num1 in list:
```

```
                list.remove(num1)
```

```
                print("List after removing:",list)
```

```
                else:
                    print("Element not found")

elif ch == 3:
    if not list:
        print("List is empty")
    else:
        print(list)

elif ch == 4:
    print("")
    break

else:
    print("Invalid choice")
```

2.1.2. Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Code:

```
student = {
    1: "Amit",
    2: "Riya",
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
```

```
8: "Sneha",
9: "Vikram",
10: "Anjali"
}

# write your code here...

def main():

    print(f"Original Dictionary: {student}")

    try:

        insert_key=int(input())

        insert_value=input()

        student[insert_key]=insert_value

    except (ValueError, EOFError):

        pass

    print(f"After Insertion: {student}")

    try:

        update_key = int(input())

        update_value = input()

        if update_key in student:

            student[update_key] = update_value

    except (ValueError, EOFError):

        pass

    print(f"After Update: {student}")

    try:

        delete_key = int(input())
```

```

        if delete_key in student:

            del student[delete_key]

    except (ValueError, EOFError):

        pass

    print(f"After Deletion: {student}")

    print("Traversing Dictionary:")

    for key, value in student.items():

        print(f"{key} : {value}")

if __name__ == "__main__":

    main()

```

2.2.1. Write a program to check whether the given element is present or not in the array of elements using linear search.

Code:

```

numbers = list(map(int, input().split()))

key = int(input())

if key in numbers:

    print(numbers.index(key))

else:

    print("Not found")

```

2.2.2. You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Code:

```

heights = list(map(int, input().split()))

```

```
tallest_player = max(heights)
```

```
print(tallest_player)
```